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Para Inf

CR Ineq. Under conditions stated earlier If δ U.E. of $\psi(\theta)$

$$V(\delta(X)) \geq \frac{(\psi'(\theta))^2}{I_X(\theta)} = \text{CRLB}$$

 δ is most eff. if $V(\delta) = \text{CRLB}$

Result: If $f(x|\theta)$; $\theta \in \Theta \subseteq \mathbb{R}$ satisfies the regularity cond^{ns} of CR Ineq. and $\hat{\theta}$ is an unbiased estimator of θ that attains CRLB, then $\hat{\theta}$ is the unique maximizer of likelihood for $L(\theta|x) = f(x|\theta)$ (under few more assumptions)

Pf:

$$\delta(x) = a(\theta) \frac{\partial}{\partial \theta} \ln f(x|\theta) + b(\theta)$$

$$\Leftrightarrow \hat{\theta} - \theta = a(\theta) \frac{\partial}{\partial \theta} \ln f(x|\theta)$$

$$\Leftrightarrow \frac{\partial}{\partial \theta} \ln f(x|\theta) = K(\theta)(\hat{\theta} - \theta); K(\theta) = \frac{1}{a(\theta)}$$

$$\frac{\partial}{\partial \theta} \ln f(x|\theta) \Big|_{\theta=\hat{\theta}} = 0 \quad \frac{\partial^2}{\partial \theta^2} \ln f(x|\theta) = K'(\theta)(\hat{\theta} - \theta) - K(\theta)$$

$$\frac{\partial^2}{\partial \theta^2} \ln f(x|\theta) \Big|_{\theta=\hat{\theta}} = -K(\hat{\theta})$$

T.S. $K(\theta) > 0 \quad \forall \theta$

$$\frac{\partial}{\partial \theta} \psi(\theta) = \mathbb{E} \left(\delta(x) \frac{\partial}{\partial \theta} \ln f(x|\theta) \right)$$

(CRLB Derivation)

$\psi(\theta) = \theta$
 $\delta(x) = \hat{\theta}$

$$1 = \mathbb{E} \left(\hat{\theta} \frac{\partial}{\partial \theta} \ln f(x|\theta) \right)$$

$$\Rightarrow \mathbb{E}[(\theta - \hat{\theta}) \frac{\partial}{\partial \theta} \ln f(X|\theta)] = 1$$

$$\text{Put } \frac{\partial}{\partial \theta} \ln f(X|\theta) = K(\theta)(\hat{\theta} - \theta)$$

$$\Rightarrow K(\theta) \mathbb{E}(\theta - \hat{\theta})^2 = 1$$

$$\Rightarrow K(\theta) > 0$$

□

Converse of the statement is false

CR Ineq for the multiparameter case (statement only)

Suppose X has pmf/pdf $f(x|\theta)$ $\theta^{k \times 1} \in \Theta \subseteq \mathbb{R}^k$ and

Θ = open in $\mathbb{R}^k$ that satisfy

conditions:

(1) Support: $\{x: f(x|\theta) > 0\}$ is free of θ

(2) $|\frac{\partial}{\partial \theta_i} \ln f(x|\theta)| < \infty \quad \forall i=1, \dots, k \quad \forall x$

(3) For any $\delta(x)$ with $\mathbb{E}|\delta(X)| < \infty$

we have $\frac{\partial}{\partial \theta_i} \int \delta(x) f(x|\theta) dx = \int \delta(x) \frac{\partial}{\partial \theta_i} f(x|\theta) dx$

Suppose $T(X) = (T_1(X), \dots, T_m(X))$ is an u.e. of $\psi(\theta)^{1 \times m}$

$= (\psi_1(\theta), \dots, \psi_m(\theta))$

with the matrix of derivatives $\phi(\theta) = \left(\left(\frac{\partial \psi_{i'}(\theta)}{\partial \theta_j} \right) \right)$

$i' = 1, \dots, m$
 $j = 1, \dots, k$

If $I_X(\theta) = \left(\left(E \left(\frac{\partial}{\partial \theta_i} \ln f(X|\theta) \right) \left(\frac{\partial}{\partial \theta_j} \ln f(X|\theta) \right) \right) \right) = \text{Fisher info matrix}$ is positive definite

then $V(T(X)) \geq \phi(\theta) I_X(\theta) \phi(\theta)^T$

[Here: $A^{k \times k} \geq B^{k \times k} \iff (A-B)$ is a non-negative definite

Special case (For pf: See Mathematical Statistics - Bickel, Doksum)

If $\psi(\theta) = \theta^{k \times 1} \Rightarrow \phi(\theta) = I_k$ So $V(T(X)) \geq I_X(\theta)^{-1}$

Ex. $X_1, \dots, X_n \sim N(\mu, \sigma^2)$ $\theta = (\mu, \sigma^2)$

u.e. of θ is $(\bar{X}, S^2) = T$

$V(T) = \begin{bmatrix} \frac{\sigma^2}{n} & 0 \\ 0 & \frac{2\sigma^4}{n-1} \end{bmatrix}$ $I_X(\theta) = \begin{bmatrix} \frac{n}{\sigma^2} & 0 \\ 0 & \frac{n}{2\sigma^4} \end{bmatrix}$

$I_X(\theta)^{-1} = \begin{bmatrix} \frac{\sigma^2}{n} & 0 \\ 0 & \frac{2\sigma^4}{n} \end{bmatrix} \leq V(T)$

We can at least say $\bar{X}$ is UMVUE of μ .

Chapman-Robbins-Kiefer Inequality (CRK)

does not assume regularity conditions needed for CRK Ineq.

Suppose X has pmf/pdf $f(x|\theta)$, $\theta \in \Theta \subseteq \mathbb{R}$

Suppose $\delta(X)$ is an u.e. of $\psi(\theta)$ with $E_\theta(\delta^2(X)) < \infty \forall \theta$

Suppose $f(x|\theta)$ is identifiable ($\theta_1 \neq \theta_2 \Rightarrow f(x|\theta_1) \neq f(x|\theta_2)$)

Suppose $\exists \phi(\neq \theta) \in \Theta$ s.t.

support $S_\theta := \{x: f(x|\theta) > 0\} \supset S_\phi := \{x: f(x|\phi) > 0\}$

Then
$$V_\theta(\delta(x)) \geq \sup_{\left\{ \phi \in \Theta: \phi \neq \theta, \right. \\ \left. S_\phi \subset S_\theta \right\}} \frac{(\psi(\theta) - \psi(\phi))^2}{V_\theta \left[\frac{f(x|\phi)}{f(x|\theta)} \right]}$$

Pf: Note $E_\theta \left\{ \frac{f(x|\phi)}{f(x|\theta)} \right\} = 1$

$$\begin{aligned} \text{Cov}_\theta(\delta(x), \frac{f(x|\phi)}{f(x|\theta)} - 1) &= E_\theta \left[\delta(x) \left\{ \frac{f(x|\phi)}{f(x|\theta)} - 1 \right\} \right] \\ &= \int \delta(x) \frac{f(x|\phi) - f(x|\theta)}{f(x|\theta)} f(x|\theta) dx \\ &= \int_{S_\phi} \delta(x) f(x|\phi) dx - \int_{S_\theta} \delta(x) f(x|\theta) dx \\ &= \psi(\phi) - \psi(\theta) \end{aligned}$$

$$\begin{aligned} (\psi(\phi) - \psi(\theta))^2 &= \text{Cov}_\theta^2(\delta(x), \frac{f(x|\phi)}{f(x|\theta)} - 1) \\ &\leq V_\theta(\delta(x)) V_\theta \frac{f(x|\phi)}{f(x|\theta)} \end{aligned}$$

$$\Rightarrow V_\theta(\delta(x)) \geq \frac{(\psi(\phi) - \psi(\theta))^2}{V_\theta \frac{f(x|\phi)}{f(x|\theta)}}$$

Take sup
 $\phi \dots$

□

Ex. $X \sim U(0, \theta); \theta > 0$

$$f(x|\theta) = \frac{1}{\theta} \mathbb{1}_{(0 < x < \theta)}$$

$$S_\theta = (0, \theta). \text{ If } \phi \in (0, \theta) \text{ then } S_\theta \supset S_\phi$$

$$\frac{f(x|\phi)}{f(x|\theta)} = \begin{cases} \frac{\theta}{\phi} & \text{if } 0 < x < \phi \text{ (i.e. } x \in S_\phi) \\ 0 & \text{if } \phi < x < \theta \text{ (i.e. } x \in S_\theta \setminus S_\phi) \end{cases}$$

$$\mathbb{E}_\theta \left[\frac{f(x|\phi)}{f(x|\theta)} \right] = 1$$

$$\begin{aligned} V_\theta \left[\frac{f(x|\phi)}{f(x|\theta)} \right] &= \mathbb{E} \left(\frac{f(x|\phi)}{f(x|\theta)} \right)^2 - 1 \\ &= \int_0^\phi \frac{\theta^2}{\phi^2} \frac{1}{\theta} dx - 1 \\ &= \frac{\theta}{\phi} - 1 \end{aligned}$$

$$\begin{aligned} \text{If } \delta \text{ u.e. of } \theta, \text{ then } V_\theta(\delta) &= \frac{(\theta - \phi)^2}{\frac{\theta}{\phi} - 1} \quad \forall \phi < \theta \\ &= \phi(\theta - \phi) \end{aligned}$$

$$\therefore V_\theta(\delta) \geq \sup_{\{\phi: \phi < \theta\}} \phi(\theta - \phi) = \frac{\theta^2}{4}$$

X is c.s.s. for θ and $2X$ u.e. of θ

$\Rightarrow 2X \Rightarrow \text{UMVUE of } \theta$

$$V(2X) = 4 \frac{\theta^2}{12} = \frac{\theta^2}{3} > \frac{\theta^2}{4} = \text{CRK Lower Bound}$$

check for $\frac{n+1}{n}X_{(n)}$ (asymptotically)

Some properties of MLE

$X \sim f(x|\theta)$ likelihood fn. $\mathcal{L}(\theta|x) = f(x|\theta)$

Defⁿ $\hat{\theta} = \text{MLE of } \theta = \underset{\theta \in \Theta}{\operatorname{argmax}} \mathcal{L}(\theta|x)$

① $\hat{\theta} \in \Theta$ up to (no possibility of having absurd est.)
Not guaranteed by MLE

② [Invariance Property] (Zehora)

Let $\mathcal{L}(\theta|x) = \text{likelihood fn.}$ and $h: \Theta \xrightarrow{\subseteq \mathbb{R}^k} \Lambda \subseteq \mathbb{R}^p$

If $\hat{\theta}$ is an MLE of θ , then $h(\hat{\theta})$ is an MLE of $h(\theta)$ "Range of h "

Fix $\lambda \in \Lambda$ and define $\Theta_\lambda = \{\theta \in \Theta: h(\theta) = \lambda\}$

Suppose if h 1-1 fn.

$$\lambda = h(\theta) \Leftrightarrow \theta = h^{-1}(\lambda)$$

$$L(\theta|x) = L(h^{-1}(\lambda)|x) = L^*(\lambda|x) = \text{induced likelihood fn.}$$

If h is not one-one $h^{-1}(\lambda) = \{\theta_1, \theta_2, \dots\}$ (some set)

We can define the induced likelihood fn.

$$L^*(\lambda|x) = \max_{\theta \in \mathcal{H}_\lambda} L(\theta|x)$$

MLE of $\lambda = h(\theta)$ means $\operatorname{argmax}_{\lambda \in \Lambda} L^*(\lambda|x)$

Let $\hat{\theta}$ = an MLE of θ and let $\hat{\lambda} = h(\hat{\theta}) \in \Lambda$

$$\text{T.S. } L^*(\hat{\lambda}|x) = \max_{\lambda \in \Lambda} L^*(\lambda|x)$$

$$\begin{aligned} \text{Note } \hat{\theta} \in \mathcal{H}_{\hat{\lambda}} \quad \text{Note } L^*(\hat{\lambda}|x) &= \max_{\theta \in \mathcal{H}_{\hat{\lambda}}} L(\theta|x) \\ &\geq L(\hat{\theta}|x) \quad (\because \hat{\theta} \in \mathcal{H}_{\hat{\lambda}}) \end{aligned}$$

$$\begin{aligned} L(\hat{\theta}|x) &\leq L^*(\hat{\lambda}|x) \leq \max_{\lambda \in \Lambda} L^*(\lambda|x) = \max_{\lambda \in \Lambda} \max_{\theta \in \mathcal{H}_\lambda} L(\theta|x) \\ &\quad \text{(from *)} \quad \underbrace{\hspace{10em}}_{\text{equality holds}} \quad = \max_{\theta \in \mathcal{H}} L(\theta|x) \\ &\quad \quad \quad = L(\hat{\theta}|x) \end{aligned}$$

(3) MLE (if unique) is always a fn of sufficient statistic
(if not unique) $\exists$ an MLE which is a fn. of sufficient stat.

Ex. $X_1, \dots, X_n \stackrel{iid}{\sim} U(\theta - 1/2, \theta + 1/2)$

$(X_{(1)}, X_{(n)})$ suff stat

$$\hat{\theta}_{MLE} = [X_{(n)} - 1/2, X_{(1)} + 1/2]$$

$$\hat{\theta} = \alpha(X_{(n)} - 1/2) + (1-\alpha)(X_{(1)} + 1/2) \quad \forall \alpha \in [0, 1]$$

$$\alpha = \sin^2 X_1$$

$$1-\alpha = \cos^2 X_1$$